

Empirical Risk and Empirical Risk Minimization - brief recap

Let $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$ be a training sample, let $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$ be a model class, and let $\ell(y, z)$ be a loss function.

- The **empirical risk** of f_θ is the average loss on the observed sample:

$$\hat{R}_n(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_\theta(x_i)).$$

- An **empirical risk minimizer** is any parameter

$$\hat{\theta} \in \arg \min_{\theta \in \Theta} \hat{R}_n(\theta).$$

The empirical risk depends on the sample, so $\hat{\theta}$ is itself random.

- When the empirical risk is a function of d variables, its first derivative is called the **gradient** and is defined by

$$\nabla \hat{R}_n(\theta) = \begin{pmatrix} \frac{\partial \hat{R}_n(\theta)}{\partial \theta_1} \\ \vdots \\ \frac{\partial \hat{R}_n(\theta)}{\partial \theta_d} \end{pmatrix} \in \mathbb{R}^d$$

- When the empirical risk is a function of d variables, its second derivative is called the **Hessian** and is defined by the symmetric matrix

$$\nabla^2 \hat{R}_n(\theta) = \begin{pmatrix} \frac{\partial^2 \hat{R}_n(\theta)}{\partial \theta_1^2} & \cdots & \frac{\partial^2 \hat{R}_n(\theta)}{\partial \theta_1 \partial \theta_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 \hat{R}_n(\theta)}{\partial \theta_d \partial \theta_1} & \cdots & \frac{\partial^2 \hat{R}_n(\theta)}{\partial \theta_d^2} \end{pmatrix} \in \mathcal{M}_d(\mathbb{R})$$

In this case, the empirical risk is strictly convex if the Hessian is positive definite, ensuring a unique minimizer found by solving $\nabla \hat{R}_n(\theta) = 0$.

Model setup

Consider two parameters $w_0, b_0 \in \mathbb{R}$ and observations $(x_i, y_i) \in \mathbb{R}^2$ such that

$$y_i = w_0 x_i + b_0 + \sigma \varepsilon_i + \Delta \delta_i$$

where $(\varepsilon_i)_{i=1}^n$ are iid random errors following the standard distribution $\mathcal{N}(0, 1)$ and $(\delta_i)_{i=1}^n$ are iid Bernoulli random variables with parameter $p \in [0, 1]$. Define the following quantities

$$\begin{aligned} S_1 &= \frac{1}{n} \sum_{i=1}^n x_i, & S_2 &= \frac{1}{n} \sum_{i=1}^n x_i^2, & T_1 &= \frac{1}{n} \sum_{i=1}^n y_i, & T_2 &= \frac{1}{n} \sum_{i=1}^n x_i y_i, \\ E_1 &= \frac{1}{n} \sum_{i=1}^n \varepsilon_i, & E_2 &= \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i, & D_1 &= \frac{1}{n} \sum_{i=1}^n \delta_i, & D_2 &= \frac{1}{n} \sum_{i=1}^n x_i \delta_i, \end{aligned}$$

and assume all x_i are distinct with $S_2 > 0$. Throughout, we use the quadratic loss

$$\ell(y, z) = \frac{1}{2} (y - z)^2.$$

Exercise 1. Linear model on $\mathbb{R}$

In this exercise we will consider the family of linear models $\Theta = \mathbb{R}$ and $\mathcal{F} = \{f_w : x \mapsto wx \mid w \in \Theta\}$.

- Write down the empirical risk $\hat{R}_n$ for this model and the loss ℓ . How many variables does it have?
- Compute the first and second derivatives of $\hat{R}_n$. Explain why solving $\hat{R}'_n = 0$ is sufficient to obtain the unique ERM.
- Compute the ERM $\hat{w}$ and the corresponding minimum value of the empirical risk. Express it as a function of the data quantities S_2, T_2 .

4. Express T_2 as a function of w_0, b_0, σ, Δ . In the noiseless setting where $\sigma = 0$ and $\Delta = 0$, analyze the behavior of $\hat{w}$ as b_0 varies. How do the magnitudes of the x_i impact the estimator?
5. When $b_0 = 0$, $\Delta = 0$ and $\sigma > 0$, what is the expectation and variance of $\hat{w}$?
6. When $b_0 = 0$, $\sigma = 0$ and $\Delta > 0$, what is the expectation of $\hat{w}$? How do outliers impact the estimator?

Exercise 2. Affine model on $\mathbb{R}$

In this exercise we will consider the family of affine models $\Theta = \mathbb{R}^2$ and $\mathcal{F} = \{f_{w,b} : x \mapsto wx + b \mid (w, b) \in \Theta\}$.

1. Write down the empirical risk $\hat{R}_n$ for this model and the loss ℓ . How many variables does it have?
2. Compute the gradient $\nabla \hat{R}_n \in \mathbb{R}^2$ and the Hessian $\nabla^2 \hat{R}_n \in \mathcal{M}_{2,2}(\mathbb{R})$.
3. Using the Cauchy-Schwartz inequality below, valid for distinct x_i ,

$$\left(\sum_{i=1}^n x_i \right)^2 < n \sum_{i=1}^n x_i^2$$

show that $\text{Tr}(\nabla^2 \hat{R}_n) \geq 0$ and $\det(\nabla^2 \hat{R}_n) > 0$. Conclude that $\hat{R}_n$ is strictly convex and setting the gradient to 0 is sufficient to obtain the unique ERM.

4. Express the ERM $(\hat{w}, \hat{b})$ as a function of the data quantities S_1, S_2, T_1, T_2 .
5. Express T_1 and T_2 as functions of w_0, b_0, σ, Δ . In the noiseless setting where $\sigma = 0$ and $\Delta = 0$, analyze the behavior of $(\hat{w}, \hat{b})$ as w_0 and b_0 vary. How does this differ from the linear model?
6. When $\Delta = 0$ and $\sigma > 0$, what are the expectations and variances of $\hat{w}$ and $\hat{b}$?
7. When $\sigma = 0$ and $\Delta > 0$, what are the expectations of $\hat{w}$ and $\hat{b}$? How do outliers impact the estimators? How does this differ from the linear model?